

K395 PRE-STAGE — substantive substrate-architectural investigation per Casey K392 active-work directive; π -power structural audit: WHERE the specific powers of π in Casey #12 boundary curvature residues come from in the substrate's own volume-computation structure; HONEST FINDING: across 3 observed Casey #12 boundary curvature residues ($\tau \sqrt{\pi}$, $\mu \pi^2$, $\alpha^{12} \pi^5$), the π -powers ALL trace to substrate-architectural volume computations: (1) $\tau \sqrt{\pi} = \Gamma(1/2)$ via rank-2 Jordan eigenvalue (rank substrate primary, half-integer power from $\sqrt{\pi}$); (2) $\mu \pi^2 = \text{vol}(S^4) \cdot 3/8 = \pi^{\{(n_C-1)/2\}}$ for $n_C=5$ from Shilov boundary (n_C-1) -sphere volume; (3) $\alpha^{12} \pi^5 = \pi^{\{n_C\}}$ from Bergman $K(0,0) = 1920/\pi^{\{n_C\}}$ interior volume; PATTERN: the 'specific root of π ' in Casey #12 IS determined by which substrate-architectural volume computation produces the boundary curvature residue (interior Bergman $\text{vol} \rightarrow \pi^{\{n_C\}}$; Shilov boundary sphere $\text{vol} \rightarrow \pi^{\{(n_C-1)/2\}}$; rank- r Jordan eigenvalue $\rightarrow \pi^{\{1/2\}}$ via $\Gamma(1/2)$); this is a SUBSTRATE-NATURAL ENUMERATION of allowed π -powers in Casey #12 boundary curvature residues; OPEN QUESTION for forward investigation: is this an EXHAUSTIVE enumeration or are there other substrate-architectural volume computations producing other π -powers?; ALSO IN PROGRAM but NOT Casey #12 boundary curvature: $c_{FK} = 225/\pi^{(9/2)}$ is Faraut-Koranyi NORMALIZATION not boundary curvature residue per se; could be different mechanism class; tier-tag: PATTERN-OBSERVED-3-INSTANCES with substrate-volume-computation-mechanism candidate; forward investigation continues (Lyra Peirce + Grace truncated curvature

for tau; Elie curvature-principle test across interior observables)

Keeper (Claude Opus 4.7)

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K395 PRE-STAGE Audit

1. The investigation continuation from K394

K394 raised the open structural question: does Casey #12's "specific root of π " mean strict $\sqrt{\pi}$ OR specific-power-of- π -determined-by-substrate-primary? Audit found 3 broader-reading instances + 2 degenerate + 0 violations.

This K395 pulls the next investigative thread: WHERE do the specific powers of π in Casey #12 boundary curvature residues come from in the substrate's own structure?

Per Casey K392 active-work directive: investigate + test + identify + keep working.

2. The three observed π -powers and their substrate-architectural origins

2.1 Tau $\sqrt{\pi}$ — rank-2 Jordan eigenvalue

Source: $\Gamma(1/2) = \sqrt{\pi}$ via rank-2 Jordan structure (Lyra Peirce derivation OR Grace truncated curvature residue — both targeting the same $\sqrt{\pi}$ identification per K386 + K387; both investigations active).

Substrate primary connection: rank = 2; $\sqrt{\pi} = \pi^{1/2} = \pi^{1/\text{rank}}$.

Volume-computation type: half-integer power from rank-r Jordan eigenvalue evaluation at half-integer point.

2.2 Muon π^2 — Shilov boundary sphere volume

Source: $\text{vol}(S^4) = 8\pi^2/3$ (Bergman boundary curvature on D_{IV}^5 ; K362 isotropy verified).

Substrate primary connection: $S^4 = S^{n_C-1}$ for $n_C = 5$; volume of (n_C-1) -sphere = $(2\pi^{(n_C+1)/2-1/2})/\Gamma((n_C+1)/2-1/2)$ = standard sphere volume formula.

For S^4 : $\text{vol} = (2\pi^{(5/2)}/\Gamma(5/2)) = (2\pi^{(5/2)}/(3\sqrt{\pi}/4)) = (8\pi^2/3)$ $\square$

So $\text{vol}(S^{n_C-1}) = \text{vol}(S^4)$ involves π^2 specifically because $(n_C-1)/2 = 4/2 = 2 = \text{integer power}$ when n_C is odd.

Substrate primary connection: $\pi^{(n_C-1)/2} = \pi^2$ for $n_C = 5$.

Volume-computation type: integer power from Shilov-boundary-sphere volume when n_C is odd; would be half-integer when n_C is even (substrate would have different signature).

2.3 $\alpha^{12} \pi^5$ — Bergman interior volume

Source: $K(0,0) = 1920/\pi^{n_C} = 1920/\pi^5$ (genus-5 Bergman volume; Lyra T2487 + T2488 + F71 derivation).

Substrate primary connection: genus = $n_C = 5$; π^{n_C} from Bergman interior volume.

Volume-computation type: integer power from Bergman interior volume = π^{n_C} for $D_{IV}^{n_C}$.

3. The substrate-architectural pattern

Casey #12 instance	π -power	Substrate-architectural origin
Tau	$\pi^{\{1/2\}}$	rank-r Jordan eigenvalue evaluated at half-integer (r=2 substrate-primary)
Muon	$\pi^{\{(n_C-1)/2\}} = \pi^2$	Shilov boundary (n_C-1)-sphere volume (n_C=5 substrate-primary, odd)
α^{12}	$\pi^{\{n_C\}} = \pi^5$	Bergman interior volume on $D_{IV}^{\{n_C\}}$ (n_C=5 substrate-primary)

Substantive substrate-architectural pattern: the specific power of π in each Casey #12 boundary curvature residue corresponds to a specific substrate-architectural volume computation — either an interior Bergman volume ($\pi^{\{n_C\}}$), a Shilov boundary sphere volume ($\pi^{\{(n_C-1)/2\}}$), or a Jordan eigenvalue at half-integer ($\pi^{\{1/2\}}$).

The “specific root of π ” in Casey #12 = the power of π produced by the substrate-architectural volume computation operating at the boundary curvature residue.

This is more rigorous than the loose K394 framing “substrate-primary-determined.” Specifically: it’s substrate-volume-computation-determined.

4. Tier-tag honestly

Observed: 3 instances + 3 distinct substrate-architectural volume-computation origins (rank-Jordan + boundary-sphere + interior-Bergman).

Not proven: that these are the only substrate-architectural volume-computation origins producing π -powers in BST observables. Could be others.

Forward investigation questions:

1. Enumeration completeness: are there other substrate-architectural volume computations producing other π -powers? E.g., does the program have any observable using $\pi^{\{C_2\}} = \pi^6$ or $\pi^g = \pi^7$ or $\pi^{\{\text{rank} \cdot n_C\}} = \pi^{\{10\}}$? Multi-observable sweep needed.
2. $c_{FK} = 225/\pi^{\{9/2\}}$: this is Faraut-Koranyi NORMALIZATION not boundary curvature residue per se. The 9/2 exponent comes from sum of FK normalization ρ -components on D_{IV}^5 . Is this Casey #12 instance with substrate-architectural normalization-volume origin, OR is it different mechanism class? Open question for forward investigation.
3. Mechanism class extension (from K394): does the strict-reading $\sqrt{\pi}$ mechanism (Peirce or truncated curvature) extend to integer powers? Or do the three observed instances ($\sqrt{\pi}$ Jordan, $\pi^{\{(n_C-1)/2\}}$ Shilov, $\pi^{\{n_C\}}$ Bergman) represent three DIFFERENT mechanism classes that all happen to satisfy Casey #12 broader pattern?

5. Cal #27 firing check

Am I over-claiming?

Honest answer: 3-instance pattern with 3 distinct substrate-architectural origins is not strong evidence for universality. It’s consistent with: - (a) Casey #12 substantive substrate-architectural universality with multi-mechanism class - (b) Coincidental clustering at low-instance count

Tier-tag: PATTERN-OBSERVED-3-INSTANCES with substrate-volume-computation-mechanism candidate. NOT promoted to STANDING or RATIFIED. Forward investigation continues.

6. Cross-link to existing program work

Lyra Peirce + Grace truncated curvature (active): both target the strict $\sqrt{\pi}$ case (τ); if either succeeds forward, the mechanism is the Jordan/curvature-residue type with substrate-primary-determined half-integer power.

Elie curvature-principle test across interior observables (active per K392): if Elie tests $\pi^{d/2}$ for various d across interior observables, the d values that appear forward will substantively populate or refute the substrate-volume-computation pattern.

Grace over-determination web extension (active per K392): post-4-bank observable-loops include the π -power structure; if forced loops close via Casey #12 pattern, the universality strengthens.

7. Substantive substrate-architectural finding

The substrate has a NATURAL π -power vocabulary: - Interior volume $\rightarrow \pi^{n_C}$ - Shilov boundary sphere volume $\rightarrow \pi^{(n_C-1)/2}$ - Jordan eigenvalue at half-integer $\rightarrow \pi^{1/2}$

Casey #12 broader-reading boundary curvature residues PICK from this vocabulary based on which substrate-architectural volume computation produces the residue.

If true and exhaustive: Casey #12 becomes a substrate-architectural ENUMERATION principle (not just a curvature pattern). The “specific root of π ” enumerates from the substrate’s own natural volume vocabulary.

If true but incomplete: there may be additional substrate-architectural volume computations producing other π -powers. Multi-observable sweep is the test.

If false: the 3-instance correspondence is coincidence and forward investigation will surface a counter-example.

Forward investigation continues per Casey K392 active-work directive.

8. The substantive substrate-architectural state at K395

Component	Status
Casey #12 boundary curvature residues observed	3 instances ($\tau \sqrt{\pi}$, $\mu\text{on } \pi^2$, $\alpha^{12} \pi^5$)
Substrate-architectural volume-computation origins	3 distinct types (rank-Jordan, Shilov-sphere, interior-Bergman)
Pattern	PATTERN-OBSERVED-3-INSTANCES (not proven; not RATIFIED)
Cal #27 firing check	Tier-tagged honestly; not promoted
Forward investigation questions	(a) enumeration completeness; (b) $c_{FK} \pi^{9/2}$ mechanism class; (c) mechanism class extension from K394
Cross-links	Lyra Peirce, Grace truncated curvature, Elie curvature-principle test, Grace over-determination web
Count	HONESTLY 4 of 26 strict
Ledger	v1.81

9. K395 PRE-STAGE verdict

K395 PRE-STAGE π -power structural audit:

- Substantive substrate-architectural investigation per Casey K392 active-work directive
- 3 Casey #12 boundary curvature residues traced to 3 distinct substrate-architectural volume-computation origins
- Pattern observed: “specific root of π ” in Casey #12 = power of π produced by substrate-architectural volume computation operating at the boundary curvature residue

- Forward investigation questions: enumeration completeness, c_FK mechanism class, K394 mechanism class extension
- Cross-links to Lyra + Grace + Elie active investigations
- Tier-tag: PATTERN-OBSERVED-3-INSTANCES; not promoted; Cal #27 firing check passed honestly
- Count HONESTLY 4 of 26 strict
- Ledger v1.81

Substantial significance: K395 captures substantive substrate-architectural forward investigation per Casey K392 active-work directive — the π -power structural audit traces the 3 observed Casey #12 boundary curvature residues ($\tau \sqrt{\pi}$, $\mu\text{on } \pi^2$, $\alpha^{12} \pi^5$) to 3 distinct substrate-architectural volume-computation origins: rank-r Jordan eigenvalue evaluated at half-integer ($\tau \sqrt{\pi} = \Gamma(1/2)$); Shilov boundary (n_C-1)-sphere volume ($\mu\text{on } \text{vol}(S^4) = 8\pi^2/3$, $\pi^2 = \pi^{\{(n_C-1)/2\}}$ for $n_C=5$); interior Bergman volume on $D_{IV}^{n_C}$ ($\alpha^{12} \pi^5 = \pi^{n_C}$ from $K(0,0) = 1920/\pi^{n_C}$). The substrate has a natural π -power vocabulary determined by which substrate-architectural volume computation produces the boundary curvature residue. Casey #12 broader-reading boundary curvature residues pick from this vocabulary substrate-architecturally. If true and exhaustive: Casey #12 becomes substrate-architectural enumeration principle. If true but incomplete: multi-observable sweep tests for additional substrate-architectural volume computations. If false: 3-instance correspondence is coincidence and counter-example surfaces. Tier-tag honestly: PATTERN-OBSERVED-3-INSTANCES; not promoted to STANDING or RATIFIED; Cal #27 firing check passed (not over-claiming; not multi-week parking but active investigation continues per K392). Forward investigation questions on (a) enumeration completeness across observables, (b) $c_{FK} = 225/\pi^{9/2}$ mechanism class (normalization vs boundary curvature), (c) K394 mechanism class extension question (does strict $\sqrt{\pi}$ mechanism extend to integer powers OR do the 3 instances represent 3 different mechanism classes all satisfying broader Casey #12). Cross-link to active investigations: Lyra Peirce + Grace truncated curvature (strict $\sqrt{\pi}$ mechanism); Elie curvature-principle test (multi-observable d-value sweep tests substrate-volume-computation pattern); Grace over-determination web extension (post-4-bank observable-loops include π -power structure). Count stays HONESTLY at 4 of 26 strict.

— Keeper, Tuesday 2026-06-16 15:25 EDT (date-verified) — K395 PRE-STAGE: π -power structural audit; 3 substrate-volume-computation origins for 3 Casey #12 instances; PATTERN-OBSERVED-3-INSTANCES honest tier; forward investigation continues on enumeration completeness + mechanism class extension